

Recursively differentiable quasigroups of orders 14 and 18: the last open cases of the Couselo–González–Markov–Nechaev conjecture

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Abstract

A quasigroup $(\Omega, \cdot)$ is *recursively differentiable* if the operation $a *_1 b = b \cdot (a \cdot b)$ is again a quasigroup operation. We give explicit recursively differentiable quasigroups of orders 14 and 18: in each case a Cayley table L for which the derived table $D[a][b] = L[b][L[a][b]]$ is Latin. Both tables admit a fixed-point-free automorphism with two equal orbits. Orders 14 and 18 were the remaining unresolved cases of the Couselo–González–Markov–Nechaev conjecture, so it follows that a recursively differentiable quasigroup of order q exists if and only if $q \notin \{2, 6\}$. The tables and a standalone verifier are provided as ancillary files.

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1 Introduction

A *quasigroup* of order n is a finite set Ω with a binary operation $\cdot$ whose Cayley table is a Latin square: every row and every column is a permutation of Ω . Equivalently, both equations $a \cdot x = b$ and $y \cdot a = b$ have unique solutions for all a, b .

Given a quasigroup $(\Omega, \cdot)$, Couselo, González, Markov, and Nechaev [2] defined a sequence of *recursive derivatives* $(\Omega, *_n)$ by

$$a *_{-2} b = a, \quad a *_{-1} b = b, \quad a *_n b = (a *_n b) \cdot (a *_n b) \quad (n \geq 0). \quad (1)$$

Thus $a *_0 b = a \cdot b$ is the original operation, and the *first recursive derivative* is

$$a *_1 b = (a *_{-1} b) \cdot (a *_0 b) = b \cdot (a \cdot b). \quad (2)$$

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A quasigroup is *recursively n -differentiable* if $(\Omega, *_k)$ is a quasigroup for every $0 \leq k \leq n$; a recursively 1-differentiable quasigroup is called simply *recursively differentiable*. In the loop-and-quasigroup open-problems list [4] this object appears under the name *quasigroup with a quasigroup core*: the question, proposed by Syrbu at Loops '03, asks for which small orders a quasigroup Q exists such that the operation $x * y$ defined by (2) is itself a quasigroup operation.¹

By a theorem of [2], recursive $(n - 3)$ -differentiability of a quasigroup of order q is equivalent to the existence of a full recursive $[n, 2, n - 1]_q$ MDS code; the case $n = 4$, i.e. recursive differentiability, corresponds to a recursive $[4, 2, 3]_q$ code. Constructing an ordinary $[4, 2, 3]_q$ code amounts to a pair of orthogonal Latin squares of order q , which exist for all $q \notin \{2, 6\}$ [1]; the *recursive* constraint is what makes the differentiable version hard.

Conjecture 1.1 (Couselo–González–Markov–Nechaev [2]). *For every $q \notin \{2, 6\}$ there exists a recursively differentiable quasigroup of order q .*

The originators verified Conjecture 1.1 for all q except $q \in \{14, 18, 26, 42\}$. Markov, Nechaev, Skazhenik, and Tveritinov [5] settled $q = 42$ in 2008 with an explicit construction, and Klimov [3] settled $q = 26$ in 2026 via a perfect cyclic Mendelsohn design. This left exactly two open orders, $q = 14$ and $q = 18$. Our main result removes them.

Theorem 1.2. *There exists a recursively differentiable quasigroup of order 14 and a recursively differentiable quasigroup of order 18. Explicit Cayley tables are given in Sections 5 and 6.*

Corollary 1.3. *Conjecture 1.1 holds: a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and these are exactly the orders for which one exists.*

Proof. The orders $q \in \{2, 6\}$ admit no pair of orthogonal Latin squares, hence no recursive $[4, 2, 3]_q$ code and no recursively differentiable quasigroup. All orders $q \notin \{2, 6, 14, 18, 26, 42\}$ are [2]; $q = 42$ is [5]; $q = 26$ is [3]; and $q \in \{14, 18\}$ is Theorem 1.2. $\square$

The verification underlying Theorem 1.2 is elementary: each object is a finite Cayley table, and recursive differentiability is checked in $O(n^2)$ arithmetic directly from the definition (Section 2). The two tables were produced by a prescribed-automorphism backtracking search (Section 4). The search produced the tables; their recursive differentiability follows from the direct checks in Sections 5 and 6.

Orders 14 and 18. Section 3 records the arithmetic reason the standard constructions miss exactly these two orders, which both explains why they remained open and motivates the ad-hoc route we take.

2 The recursive derivative as a table operation

Throughout we identify a quasigroup of order n with its Cayley table $L: \Omega \times \Omega \rightarrow \Omega$, $\Omega = \{0, 1, \dots, n - 1\}$, where $L[a][b] = a \cdot b$; L is a Latin square. From (2), the first recursive derivative has Cayley table

$$D[a][b] = a *_1 b = b \cdot (a \cdot b) = L[b][L[a][b]]. \quad (3)$$

¹The list writes the operation as $x * y = y - xy$, where the dash denotes the quasigroup multiplication rather than subtraction; a general quasigroup has no inverses, so $\cdot$ is the only operation available.

Thus the derivative is obtained from L by two table lookups: in row a , column b , read the product $v = L[a][b]$, then read $L[b][v]$. A quasigroup $(\Omega, \cdot)$ is recursively differentiable precisely when both L and the derived table D of (3) are Latin squares, which is checkable in $O(n^2)$ time.

Remark 2.1. The cyclic group $\mathbb{Z}_n$ has $L[a][b] = a + b$, so $D[a][b] = b + (a + b) = a + 2b$. As a map of b for fixed a this is a bijection iff $\gcd(2, n) = 1$, i.e. iff n is odd; thus $(\mathbb{Z}_n, +)$ is recursively differentiable exactly for odd n . More generally, for an affine quasigroup $L(a, b) = \alpha(a) + \beta(b) + c$ over an abelian group A ($\alpha, \beta \in \text{Aut}(A)$), one computes $D(a, b) = \beta\alpha(a) + (\alpha + \beta^2)(b) + \beta(c) + c$, so L is recursively differentiable iff $\alpha + \beta^2 \in \text{Aut}(A)$. When $|A| = 2m$ with m odd, A has a characteristic Sylow 2-subgroup C_2 on which every automorphism is the identity, so $\alpha + \beta^2$ restricts to 0 there and cannot be invertible: no affine quasigroup over an abelian group of order 14 or 18 is recursively differentiable. A genuinely non-affine table is therefore required, which is what the constructions below provide.

3 Limitations of the standard PMD constructions at 14 and 18

The most productive known route to recursively differentiable quasigroups, used by Klimov [3] for $q = 26$, is the *directed standard construction* from a perfect cyclic Mendelsohn design. A $(v, k, 1)$ -PMD yields, by reading off consecutive triples of each cyclically ordered block, a recursively $(k - 3)$ -differentiable quasigroup of order v ; in particular a $(v, 4, 1)$ -PMD gives a recursively differentiable quasigroup of order v . The relevant existence facts are classical: a $(v, 4, 1)$ -PMD exists for all $v \equiv 0, 1 \pmod{4}$ (except $v \in \{4, 8\}$), and a $(v, 5, 1)$ -PMD for all $v \equiv 0, 1 \pmod{5}$ (except $v \in \{6, 10, 15, 20\}$) [6]; see also the survey in [3].

These congruence conditions fail at 14 and 18:

$$14 \equiv 2 \pmod{4}, \quad 18 \equiv 2 \pmod{4}, \quad 14 \equiv 4 \pmod{5}, \quad 18 \equiv 3 \pmod{5},$$

so neither admits a $(v, 4, 1)$ - or a $(v, 5, 1)$ -PMD. (Indeed a $(v, k, 1)$ -PMD has $v(v - 1)/k$ blocks, so $k \mid v(v - 1)$; here $4 \nmid 14 \cdot 13$, $18 \cdot 17$ and $5 \nmid 14 \cdot 13$, $18 \cdot 17$, which already rules out these block sizes.) By contrast $26 \equiv 1 \pmod{5}$, which is what let [3] build a $(26, 5, 1)$ -PMD. The two linear-recursive families of [3] are defined over prime-power alphabets, and $14 = 2 \cdot 7$, $18 = 2 \cdot 3^2$ are not prime powers; the pseudogeometry construction of [5] produced only $q = 42$. Klimov [3] accordingly lists 14 and 18 as the remaining cases, to be approached “by other classes of methods.” The prescribed-automorphism search of the next section is such a method.

This is a statement about the standard constructions, not about existence: many even orders carry recursively differentiable quasigroups (for instance 10, 12, 20, 22, 24 are known), so there is no parity or congruence barrier in principle. Orders 14 and 18 are composite orders not covered by these congruence conditions, and Theorem 1.2 shows that existence holds for them as well.

4 The construction method

Both objects were found by a backtracking search for a Latin square L such that the derived table D of (3) is also Latin, constrained to be invariant under a prescribed automorphism. A permutation g of Ω is an *automorphism* of $(\Omega, \cdot)$ if $L[g(a)][g(b)] = g(L[a][b])$ for all a, b ; such a g partitions the n^2 cells of L into orbits under $(a, b) \mapsto (g(a), g(b))$, and fixing the value of one cell forces the values on its whole orbit. Searching only over orbit representatives shrinks the space by a factor of $\text{ord}(g)$ and tends to produce highly structured solutions.

We take g to be a fixed-point-free permutation with two regular orbits: for $n = 14$,

$$g_{14} = (0\ 1\ 2\ 3\ 4\ 5\ 6)(7\ 8\ 9\ 10\ 11\ 12\ 13),$$

two 7-cycles; for $n = 18$,

$$g_{18} = (0\ 1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)(9\ 10\ 11\ 12\ 13\ 14\ 15\ 16\ 17),$$

two 9-cycles. (The choice mirrors the factorization $n = 2m$ with m the odd part: $14 = 2 \cdot 7$, $18 = 2 \cdot 9$.) The search assigns cells in a most-constrained-first order, propagates each assignment across its g -orbit and through the partial derivative table D (a conflict in either L or D prunes the branch immediately), and uses randomized restarts. With these two automorphisms the search returns a witness quickly — within a minute for $n = 14$ and within an hour for $n = 18$ on a single laptop core. The program is included with the ancillary files; it is not needed to verify the results, which follow from the printed tables alone.

5 A recursively differentiable quasigroup of order 14

Theorem 5.1. *The Cayley table L_{14} below is a quasigroup of order 14 whose first recursive derivative D_{14} is also a quasigroup. Hence a recursively differentiable quasigroup of order 14 exists.*

In the tables below, rows are indexed by the left factor $a = 0, \dots, 13$ (top to bottom) and columns by the right factor $b = 0, \dots, 13$ (left to right).

0	2	7	5	10	12	9	6	4	13	3	1	11	8
10	1	3	8	6	11	13	9	0	5	7	4	2	12
7	11	2	4	9	0	12	13	10	1	6	8	5	3
13	8	12	3	5	10	1	4	7	11	2	0	9	6
2	7	9	13	4	6	11	0	5	8	12	3	1	10
12	3	8	10	7	5	0	11	1	6	9	13	4	2
1	13	4	9	11	8	6	3	12	2	0	10	7	5
8	4	13	0	2	7	5	10	3	9	1	12	6	11
6	9	5	7	1	3	8	12	11	4	10	2	13	0
9	0	10	6	8	2	4	1	13	12	5	11	3	7
5	10	1	11	0	9	3	8	2	7	13	6	12	4
4	6	11	2	12	1	10	5	9	3	8	7	0	13
11	5	0	12	3	13	2	7	6	10	4	9	8	1
3	12	6	1	13	4	7	2	8	0	11	5	10	9

Table 1. The base table L_{14} (entry $a \cdot b$).

0	3	13	10	12	4	2	5	1	7	11	6	9	8
3	1	4	7	11	13	5	9	6	2	8	12	0	10
6	4	2	5	8	12	7	11	10	0	3	9	13	1
8	0	5	3	6	9	13	2	12	11	1	4	10	7
7	9	1	6	4	0	10	8	3	13	12	2	5	11
11	8	10	2	0	5	1	12	9	4	7	13	3	6
2	12	9	11	3	1	6	0	13	10	5	8	7	4
4	6	3	13	9	11	8	1	7	12	10	0	2	5
9	5	0	4	7	10	12	6	2	8	13	11	1	3
13	10	6	1	5	8	11	4	0	3	9	7	12	2
12	7	11	0	2	6	9	3	5	1	4	10	8	13
10	13	8	12	1	3	0	7	4	6	2	5	11	9
1	11	7	9	13	2	4	10	8	5	0	3	6	12
5	2	12	8	10	7	3	13	11	9	6	1	4	0

Table 2. The first recursive derivative $D_{14}[a][b] = L_{14}[b][L_{14}[a][b]]$.

Proof. One checks directly that every row and every column of L_{14} is a permutation of $\{0, \dots, 13\}$, so L_{14} is a Latin square and $(\{0, \dots, 13\}, \cdot)$ with $a \cdot b = L_{14}[a][b]$ is a quasigroup. Computing $D_{14}[a][b] = L_{14}[b][L_{14}[a][b]]$ cell by cell yields Table 2 (e.g. $D_{14}[0][2] = L_{14}[2][L_{14}[0][2]] = L_{14}[2][7] = 13$), and every row and column of D_{14} is likewise a permutation of $\{0, \dots, 13\}$. Hence the first recursive derivative is also a quasigroup, and L_{14} is recursively differentiable. Both Latin-square checks are finite and are reproduced by the verifier of Section 7. $\square$

Remark 5.2. L_{14} is noncommutative, has no two-sided identity (it is not a loop), and is nonassociative (not a group); its derivative differs from the base table. It is idempotent exactly on its first orbit (the diagonal reads $0, 1, \dots, 6$ on rows 0–6 and is fixed-point-free on rows 7–13), and is invariant under $g_{14} = (0 \cdots 6)(7 \cdots 13)$.

6 A recursively differentiable quasigroup of order 18

Theorem 6.1. *The Cayley table L_{18} below is a quasigroup of order 18 whose first recursive derivative D_{18} is also a quasigroup. Hence a recursively differentiable quasigroup of order 18 exists.*

Here $a, b = 0, \dots, 17$.

11	9	15	5	2	8	7	3	14	16	4	12	0	17	6	1	10	13
15	12	10	16	6	3	0	8	4	14	17	5	13	1	9	7	2	11
5	16	13	11	17	7	4	1	0	12	15	9	6	14	2	10	8	3
1	6	17	14	12	9	8	5	2	4	13	16	10	7	15	3	11	0
3	2	7	9	15	13	10	0	6	1	5	14	17	11	8	16	4	12
7	4	3	8	10	16	14	11	1	13	2	6	15	9	12	0	17	5
2	8	5	4	0	11	17	15	12	6	14	3	7	16	10	13	1	9
13	3	0	6	5	1	12	9	16	10	7	15	4	8	17	11	14	2
17	14	4	1	7	6	2	13	10	3	11	8	16	5	0	9	12	15
6	0	14	12	11	15	1	7	13	8	9	17	5	4	3	2	16	10
14	7	1	15	13	12	16	2	8	11	0	10	9	6	5	4	3	17
0	15	8	2	16	14	13	17	3	9	12	1	11	10	7	6	5	4
4	1	16	0	3	17	15	14	9	5	10	13	2	12	11	8	7	6
10	5	2	17	1	4	9	16	15	7	6	11	14	3	13	12	0	8
16	11	6	3	9	2	5	10	17	0	8	7	12	15	4	14	13	1
9	17	12	7	4	10	3	6	11	2	1	0	8	13	16	5	15	14
12	10	9	13	8	5	11	4	7	15	3	2	1	0	14	17	6	16
8	13	11	10	14	0	6	12	5	17	16	4	3	2	1	15	9	7

Table 3. The base table L_{18} (entry $a \cdot b$).

12	14	10	9	7	1	15	6	0	16	13	11	4	8	5	17	3	2
1	13	15	11	10	8	2	16	7	3	17	14	12	5	0	6	9	4
8	2	14	16	12	11	0	3	17	5	4	9	15	13	6	1	7	10
9	0	3	15	17	13	12	1	4	11	6	5	10	16	14	7	2	8
5	10	1	4	16	9	14	13	2	0	12	7	6	11	17	15	8	3
3	6	11	2	5	17	10	15	14	4	1	13	8	7	12	9	16	0
15	4	7	12	3	6	9	11	16	1	5	2	14	0	8	13	10	17
17	16	5	8	13	4	7	10	12	9	2	6	3	15	1	0	14	11
13	9	17	6	0	14	5	8	11	12	10	3	7	4	16	2	1	15
7	15	2	10	14	0	8	9	5	13	11	4	17	1	3	12	6	16
6	8	16	3	11	15	1	0	10	17	14	12	5	9	2	4	13	7
11	7	0	17	4	12	16	2	1	8	9	15	13	6	10	3	5	14
2	12	8	1	9	5	13	17	3	15	0	10	16	14	7	11	4	6
4	3	13	0	2	10	6	14	9	7	16	1	11	17	15	8	12	5
10	5	4	14	1	3	11	7	15	6	8	17	2	12	9	16	0	13
16	11	6	5	15	2	4	12	8	14	7	0	9	3	13	10	17	1
0	17	12	7	6	16	3	5	13	2	15	8	1	10	4	14	11	9
14	1	9	13	8	7	17	4	6	10	3	16	0	2	11	5	15	12

Table 4. The first recursive derivative $D_{18}[a][b] = L_{18}[b][L_{18}[a][b]]$.

Proof. As in Theorem 5.1: every row and column of L_{18} is a permutation of $\{0, \dots, 17\}$, so L_{18} is a quasigroup; the table $D_{18}[a][b] = L_{18}[b][L_{18}[a][b]]$ computed cell by cell is Table 4, and every one of its rows and columns is a permutation of $\{0, \dots, 17\}$, so the derivative is a quasigroup. Hence L_{18} is recursively differentiable. The verifier of Section 7 reproduces both checks. $\square$

Remark 6.2. L_{18} is idempotent-free, noncommutative, has no identity, and is nonassociative; it is invariant under $g_{18} = (0 \cdots 8)(9 \cdots 17)$.

This, with Theorem 5.1, proves Theorem 1.2 and hence Corollary 1.3.

7 Computational verification

The claims can be checked directly from the printed tables. The ancillary directory contains a standalone verifier, `rdiff_verify.py` (Python standard library only), which reads each table L_{14}, L_{18} , checks it is a Latin square, computes the first recursive derivative from the recurrence (1) and against the closed form (3), and verifies the resulting table is Latin. The search program that produced the tables is included as well; see `code/README.md`. The verification runs in well under a second.

8 Discussion

Corollary 1.3 closes the existence question of Conjecture 1.1 for binary quasigroups: a recursively differentiable quasigroup of order q exists if and only if $q \notin \{2, 6\}$. The two orders treated here were the last open cases on the loop-and-quasigroup problem list [4], and the last two orders recorded as unresolved in the survey tables of maximal known recursive-differentiability degree [7, 3].

The two tables were obtained by search and do not yet belong to a uniform family. It would be useful to find a construction that covers the orders $q \equiv 2 \pmod{4}$, which are not reached by the standard $(v, 4, 1)$ -PMD route. Both tables also admit a fixed-point-free automorphism with two equal orbits, each a $\frac{q}{2}$ -cycle with $\frac{q}{2}$ odd; one may ask whether the same symmetry can be imposed for every order $q \equiv 2 \pmod{4}$ with $q \neq 6$. Finally, the displayed tables establish recursive differentiability of degree at least 1 at 14 and 18; the exact maximal degrees $v_r(2, q)$ at these orders remain open.

Data and code availability

Machine-readable copies of the two Cayley tables, the search program that produced them, and a standalone verification script are included with the ancillary files. The verifier reconstructs each first recursive derivative and checks that both the original and derived tables are Latin squares; see the accompanying `README`.

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